Intermediate_Elective_2

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1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(NatParksPalettes)
library(RColorBrewer)

# reading .xlxs files
library(readxl)

# wrap axis labels
library(scales)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# read in vegatation data
veg <- read_csv(here("data", "veg.csv"))

# vegetation survey metadata
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

2. Problems

1. Explain your inspiration

I chose Cristina Cametti's "IKEA most popular designers" to represent total percent coverage of each vegetation species across all vernal pools in the year of 2025. The variables shown are the percent cover relative to each species, the species names, and cover types (each percent cover will be its own column)

2. PLan your figure

ViewPDF

3. Code your figure

```
# created a new object from veg
pools_explicit0_25 <- veg |>
  # filtered to only include vernal pool transects
  filter(str_detect(transect_name, "VP") &
           site == "ncos") |>
  # added new rows to dataframe for missing species and transect distances
   ↪ for each year and pool
  complete(year, transect_name, psoc, transect_distance,
           # filled missing percent cover values with 0
           fill = list(percent_cover = 0)) |>
  # grouped by year, pool, species, and cover category
  group_by(year, transect_name, psoc, cover_category) |>
  # summarized by calculating total percent cover across all quadrats
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
  # ungrouped the data fram
  ungroup() |>
```

```r
  # put remove = TRUE to remove year and transect name columns to avoid
  ↪ redundancy
  # created a column to join with metadata
  unite("year_pool", year, transect_name, remove = TRUE) |>
  # joined metadata data frame
  left_join(metadata, by = "year_pool") |>
  # calculated mean percent cover
  mutate(mean_pc = sum_pc/num_quad) |>
  # only include 2025 data
  filter(year == "2025")

# created a new object from pools_explicit_25
total_pool_coverage <- pools_explicit0_25 |>
  # grouped by species and cover category
  group_by(psoc, cover_category) |>
  # added mean_pc together across all pools
  summarize(
    n = sum(mean_pc, na.rm = TRUE)
  ) |>
  # ungrouped
  ungroup()
```

```r
# created new object from total_pool_coverage
dt <- total_pool_coverage |>
  # assigned new names to my data to easily plug into Cametti's data
  rename(designer = psoc,
    category = cover_category) |>
  # arranged in descending order
  arrange(desc(n)) |>
  # filtered out 0s
  filter(n > 5) |>
  # mutated to make the dataset values smaller
  mutate(n = sqrt(n),
  designer = factor(designer, levels = designer))

# turning columns into factors
dt <- mutate_at(dt, vars(category), as.factor)
dt <- mutate_at(dt, vars(designer), as.factor)
glimpse(dt)
```

```
Rows: 20
Columns: 3
```

```
$ designer <fct> Eleocharis_palustris, biocrust, Natural Thatch, Eryngium_vase~
$ category <fct> NATIVE COVER, OTHER COVER, THATCH, NATIVE COVER, BARE GROUND,~
$ n        <dbl> 16.631328, 15.207638, 11.974814, 8.697333, 8.142223, 8.105323~
```

```r
# Set a number of 'empty bar' to add at the end of each group
empty_bar <- 2
# Add lines to the initial dataset
to_add <- matrix(NA, empty_bar, ncol(dt))
colnames(to_add) <- colnames(dt)
dt <- rbind(dt, to_add)
dt$id <- seq(1, nrow(dt))

# Get the name and the y position of each label
label_data <- dt
number_of_bar <- nrow(label_data)
angle <- 90 - 360 * (label_data$id-0.5) /number_of_bar
label_data$hjust <- ifelse(angle < -90, 1, 0)
label_data$angle <- ifelse(angle < -90, angle+180, angle)

# palette
pal <- brewer.pal(11, "Spectral")
```

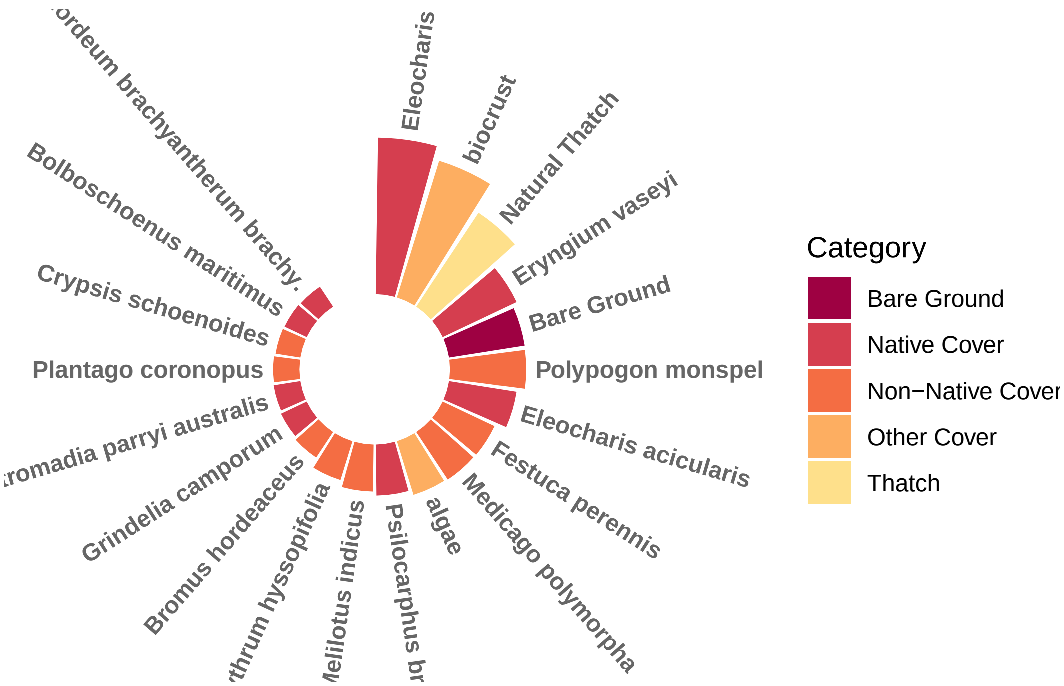
```r
figure <- ggplot(dt) +
  geom_bar(aes(x = as.factor(id), y = n, fill = category), stat = "identity")
   ↪  +
  scale_fill_manual(
    values = pal,
    name = "Category",
    labels = c("Bare Ground", "Native Cover", "Non-Native Cover", "Other
     ↪  Cover", "Thatch"),
    na.translate = FALSE
  ) +
  ylim(-8, 25) +
  theme_minimal() +
  theme(
    axis.text = element_blank(),
    axis.title = element_blank(),
    panel.grid = element_blank(),
    plot.margin = unit(rep(-1, 4), "cm")
  ) +
  labs(title = "Species Cover across Vernal Pools 2025") +
  coord_polar(start = 0) +
```

```
  geom_text(
    data = label_data,
    aes(x = id,
        # adding labels with 1 space over the column
        y = n + 1,
        # cleaning labels
        label = str_replace_all(designer, "_", " "),
        hjust = hjust),
    color = "black",
    fontface = "bold",
    alpha = 0.6,
    size = 3.0,
    angle = label_data$angle,
    inherit.aes = FALSE)
```

```
# save figure as pdf

ggsave("percentcover.png",
  plot = figure,
  width = 13,
  height = 13,
  dpi = 300
)

figure
```



4. Describe your process

Integrating my data into code that I was unfamiliar with was an extremely challenging task, it meant understanding their new code and tweaking my data to fit their parameters. The most challenging thing was altering my data in a way that made sense for the visualization as the figure required a certain amount of variables. Ultimately, this visual needed to tell a story, and I feel that I ended up showing a succinct comparison of what the vernal pools contained in the year 2025 and what types of ground cover they are. This graph is basically a bar graph, only curved for aesthetics.